



UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

CSA3023

WEB-BASED APPLICATION DEVELOPMENT

LAB 8:

AN MVC EXAMPLE WITH SERVLETS AND JSP

Prepared by:

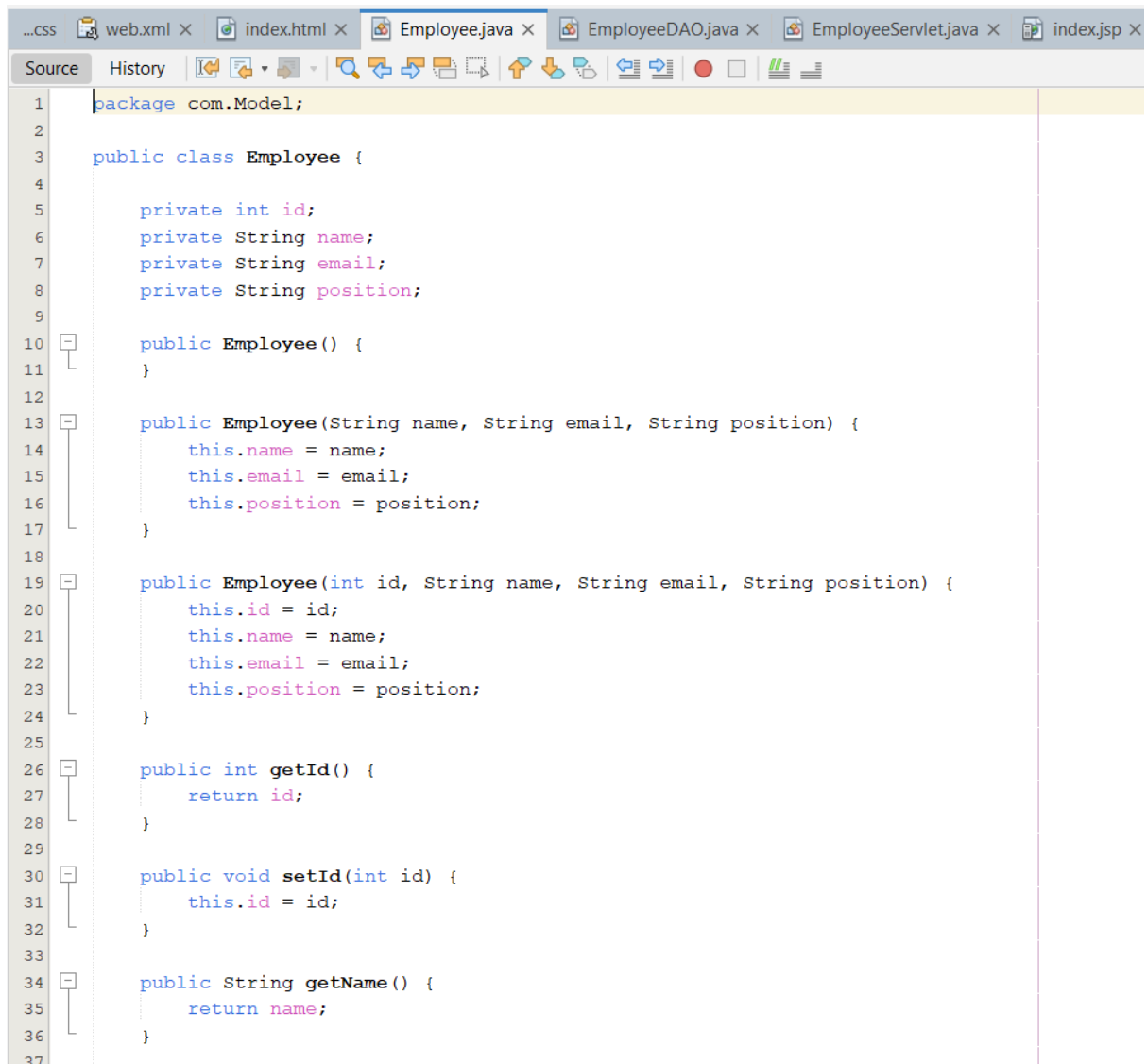
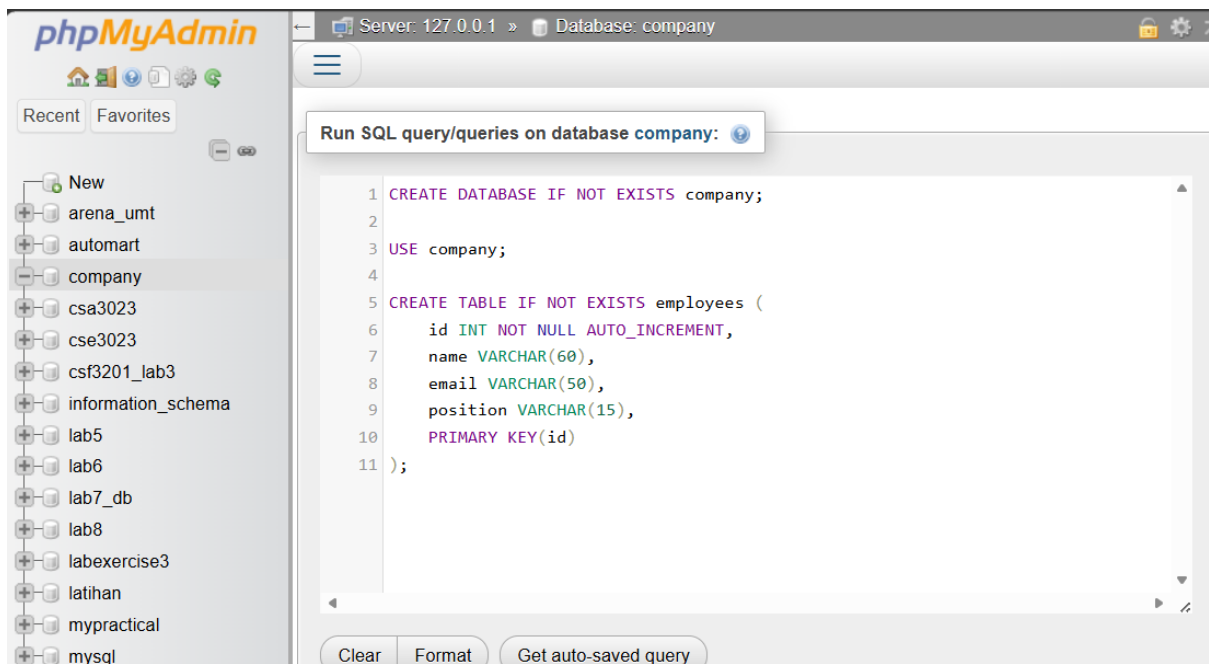
NURSYAHEERA BINTI MOHD HISHAM

Prepared for:

SIR. MOHAMAD ARIZAL SHAMSIL MAT RIFFIN

Date:

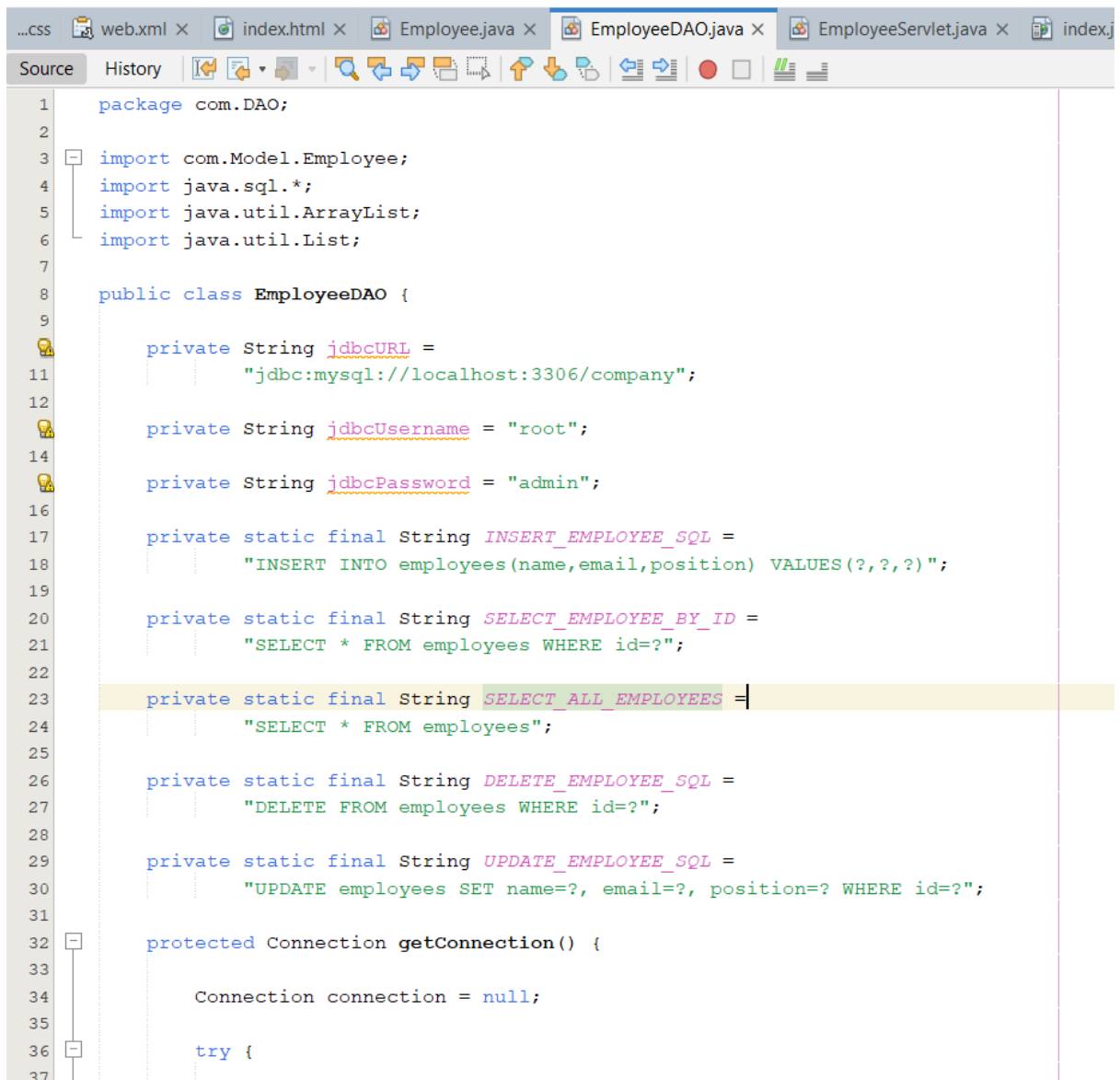
2 JUNE 2026



```

38     public void setName(String name) {
39         this.name = name;
40     }
41
42     public String getEmail() {
43         return email;
44     }
45
46     public void setEmail(String email) {
47         this.email = email;
48     }
49
50     public String getPosition() {
51         return position;
52     }
53
54     public void setPosition(String position) {
55         this.position = position;
56     }
57 }

```



The screenshot shows an IDE with several tabs open: web.xml, index.html, Employee.java, EmployeeDAO.java (active), EmployeeServlet.java, and index.j. The EmployeeDAO.java file contains the following code:

```

1  package com.DAO;
2
3  import com.Model.Employee;
4  import java.sql.*;
5  import java.util.ArrayList;
6  import java.util.List;
7
8  public class EmployeeDAO {
9
10     private String jdbcURL =
11         "jdbc:mysql://localhost:3306/company";
12
13     private String jdbcUsername = "root";
14
15     private String jdbcPassword = "admin";
16
17     private static final String INSERT_EMPLOYEE_SQL =
18         "INSERT INTO employees(name,email,position) VALUES(?,?,?)";
19
20     private static final String SELECT_EMPLOYEE_BY_ID =
21         "SELECT * FROM employees WHERE id=?";
22
23     private static final String SELECT_ALL_EMPLOYEES =
24         "SELECT * FROM employees";
25
26     private static final String DELETE_EMPLOYEE_SQL =
27         "DELETE FROM employees WHERE id=?";
28
29     private static final String UPDATE_EMPLOYEE_SQL =
30         "UPDATE employees SET name=?, email=?, position=? WHERE id=?";
31
32     protected Connection getConnection() {
33
34         Connection connection = null;
35
36         try {
37

```

```

38         Class.forName("com.mysql.cj.jdbc.Driver");
39
40         connection = DriverManager.getConnection(
41             jdbcURL,
42             jdbcUsername,
43             jdbcPassword);
44
45     } catch (Exception e) {
46         e.printStackTrace();
47     }
48
49     return connection;
50 }
51
52 public void insertEmployee(Employee employee)
53     throws SQLException {
54
55     try (Connection connection = getConnection();
56         PreparedStatement statement =
57             connection.prepareStatement(INSERT_EMPLOYEE_SQL)) {
58
59         statement.setString(1, employee.getName());
60         statement.setString(2, employee.getEmail());
61         statement.setString(3, employee.getPosition());
62
63         statement.executeUpdate();
64     }
65 }
66
67 public Employee selectEmployee(int id) {
68
69     Employee employee = null;
70
71     try (Connection connection = getConnection();
72         PreparedStatement statement =
73             connection.prepareStatement(SELECT_EMPLOYEE_BY_ID)) {
74

```

```

75         statement.setInt(1, id);
76
77         ResultSet rs = statement.executeQuery();
78
79         while (rs.next()) {
80
81             String name = rs.getString("name");
82             String email = rs.getString("email");
83             String position = rs.getString("position");
84
85             employee = new Employee(id, name, email, position);
86         }
87
88     } catch (Exception e) {
89         e.printStackTrace();
90     }
91
92     return employee;
93 }
94
95 public List<Employee> selectAllEmployees() {
96
97     List<Employee> employees = new ArrayList<>();
98
99     try (Connection connection = getConnection();
100         PreparedStatement statement =
101             connection.prepareStatement(SELECT_ALL_EMPLOYEES)) {
102
103         ResultSet rs = statement.executeQuery();
104
105         while (rs.next()) {
106
107             int id = rs.getInt("id");
108             String name = rs.getString("name");
109             String email = rs.getString("email");
110             String position = rs.getString("position");
111

```

```

112         employees.add(
113             new Employee(id, name, email, position));
114     }
115
116     } catch (Exception e) {
117         e.printStackTrace();
118     }
119
120     return employees;
121 }
122
123 public boolean updateEmployee(Employee employee)
124     throws SQLException {
125
126     boolean rowUpdated;
127
128     try (Connection connection = getConnection();
129         PreparedStatement statement =
130             connection.prepareStatement(UPDATE_EMPLOYEE_SQL)) {
131
132         statement.setString(1, employee.getName());
133         statement.setString(2, employee.getEmail());
134         statement.setString(3, employee.getPosition());
135         statement.setInt(4, employee.getId());
136
137         rowUpdated = statement.executeUpdate() > 0;
138     }
139
140     return rowUpdated;
141 }
142
143 public boolean deleteEmployee(int id)
144     throws SQLException {
145
146     boolean rowDeleted;
147
148     public boolean deleteEmployee(int id)
149         throws SQLException {
150
151         boolean rowDeleted;
152
153         try (Connection connection = getConnection();
154             PreparedStatement statement =
155                 connection.prepareStatement(DELETE_EMPLOYEE_SQL)) {
156
157             statement.setInt(1, id);
158
159             rowDeleted = statement.executeUpdate() > 0;
160         }
161
162         return rowDeleted;
163     }
164 }

```

...css web.xml × index.html × Employee.java × EmployeeDAO.java × EmployeeServlet.java × index.jsp

Source History

```
1 package com.WEB;
2
3 import com.DAO.EmployeeDAO;
4 import com.Model.Employee;
5
6 import java.io.IOException;
7 import java.sql.SQLException;
8 import java.util.List;
9
10 import javax.servlet.RequestDispatcher;
11 import javax.servlet.ServletException;
12 import javax.servlet.annotation.WebServlet;
13 import javax.servlet.http.HttpServlet;
14 import javax.servlet.http.HttpServletRequest;
15 import javax.servlet.http.HttpServletResponse;
16 @WebServlet("/")
17 public class EmployeeServlet extends HttpServlet {
18
19     private EmployeeDAO employeeDAO;
20
21     public void init() {
22         employeeDAO = new EmployeeDAO();
23     }
24
25     @Override
26     protected void doPost(HttpServletRequest request,
27                           HttpServletResponse response)
28         throws ServletException, IOException {
29
30         doGet(request, response);
31     }
32
33     @Override
34     protected void doGet(HttpServletRequest request,
35                           HttpServletResponse response)
36         throws ServletException, IOException {
37
```

```

38     String action = request.getServletPath();
39
40     try {
41         switch (action) {
42             case "/new":
43                 showNewForm(request, response);
44                 break;
45             case "/insert":
46                 insertEmployee(request, response);
47                 break;
48             case "/delete":
49                 deleteEmployee(request, response);
50                 break;
51             case "/edit":
52                 showEditForm(request, response);
53                 break;
54             case "/update":
55                 updateEmployee(request, response);
56                 break;
57             default:
58                 listEmployee(request, response);
59                 break;
60         }
61     } catch (SQLException ex) {
62         throw new ServletException(ex);
63     }
64 }
65
66 private void listEmployee(HttpServletRequest request,

```



```
75         HttpServletResponse response)
76         throws SQLException, IOException, ServletException {
77
78         List<Employee> listEmployee
79             = employeeDAO.selectAllEmployees();
80
81         request.setAttribute("listEmployee", listEmployee);
82
83         RequestDispatcher dispatcher
84             = request.getRequestDispatcher("employeeList.jsp");
85
86         dispatcher.forward(request, response);
87     }
88
89     private void showNewForm(HttpServletRequest request,
90                             HttpServletResponse response)
91         throws ServletException, IOException {
92
93         RequestDispatcher dispatcher
94             = request.getRequestDispatcher("employeeForm.jsp");
95
96         dispatcher.forward(request, response);
97     }
98
99     private void showEditForm(HttpServletRequest request,
100                              HttpServletResponse response)
101         throws SQLException, ServletException, IOException {
102
103         int id = Integer.parseInt(
104             request.getParameter("id"));
105
106         Employee existingEmployee
107             = employeeDAO.selectEmployee(id);
108
109         RequestDispatcher dispatcher
110             = request.getRequestDispatcher("employeeForm.jsp");
111     }
```

```

112         request.setAttribute(
113             "employee",
114             existingEmployee);
115
116         dispatcher.forward(request, response);
117     }
118
119     private void insertEmployee(HttpServletRequest request,
120         HttpServletResponse response)
121         throws SQLException, IOException {
122
123         String name = request.getParameter("name");
124         String email = request.getParameter("email");
125         String position = request.getParameter("position");
126
127         Employee newEmployee
128             = new Employee(name, email, position);
129
130         employeeDAO.insertEmployee(newEmployee);
131
132         response.sendRedirect("list");
133     }
134
135     private void updateEmployee(HttpServletRequest request,
136         HttpServletResponse response)
137         throws SQLException, IOException {
138
139         int id = Integer.parseInt(
140             request.getParameter("id"));
141
142         String name = request.getParameter("name");
143         String email = request.getParameter("email");
144         String position = request.getParameter("position");
145
146         Employee employee
147             = new Employee(
148             id,
149             name,
150             email,
151             position);
152
153         employeeDAO.updateEmployee(employee);
154
155         response.sendRedirect("list");
156     }
157
158     private void deleteEmployee(HttpServletRequest request,
159         HttpServletResponse response)
160         throws SQLException, IOException {
161
162         int id = Integer.parseInt(
163             request.getParameter("id"));
164
165         employeeDAO.deleteEmployee(id);
166
167         response.sendRedirect("list");
168     }
169 }

```

```
...css web.xml x index.html x Employee.java x EmployeeDAO.java x EmployeeServlet.java x index.jsp x
Source History
6
7 <title>Employee Management</title>
8
9 <link rel="stylesheet"
10 href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
11
12 </head>
13
14 <body>
15
16 <div class="container mt-5">
17
18 <h1>Employee Management MVC System</h1>
19
20 <hr>
21
22 <a href="list" class="btn btn-primary">
23 View All Employees
24 </a>
25
26 <a href="new" class="btn btn-success">
27 Add Employee
28 </a>
29
30 </div>
31
32 </body>
33 </html>
```

```
...css web.xml x index.html x Employee.java x EmployeeDAO.java x EmployeeServlet.java x index.jsp x employeeForm.jsp x
Source History
1 <%@ page contentType="text/html; charset=UTF-8" %>
2 <%@ taglib uri="http://java.sun.com/jsp/jstl/core"
3 prefix="c" %>
4
5 <!DOCTYPE html>
6
7 <html>
8 <head>
9
10 <title>Employee Form</title>
11
12 <link rel="stylesheet"
13 href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
14
15 </head>
16
17 <body>
18
19 <div class="container mt-4">
20
21 <div class="card">
22
23 <div class="card-body">
24
25 <c:if test="{employee != null}">
26 <form action="update" method="post">
27 </c:if>
28
29 <c:if test="{employee == null}">
30 <form action="insert" method="post">
31 </c:if>
32
33 <c:if test="{employee != null}">
34 <input type="hidden"
35 name="id"
36 value="{employee.id}">
37 </c:if>
```

```

38
39 <h2>
40
41 <c:if test="{employee != null}">
42   Edit Employee
43 </c:if>
44
45 <c:if test="{employee == null}">
46   Add Employee
47 </c:if>
48
49 </h2>
50
51 <div class="form-group">
52
53   <label>Name</label>
54
55   <input type="text"
56     name="name"
57     class="form-control"
58     value="{employee.name}"
59     required>
60
61 </div>
62
63 <div class="form-group">
64
65   <label>Email</label>
66
67   <input type="email"
68     name="email"
69     class="form-control"
70     value="{employee.email}"
71     required>
72
73 </div>

```

```

...css web.xml x index.html x Employee.java x EmployeeDAO.java x EmployeeServlet.java x index.jsp x employeeForm.jsp x employeeList.jsp x
Source History
<? page contentType="text/html; charset=UTF-8" ?>
2 <? taglib uri="http://java.sun.com/jsp/jstl/core"
3   prefix="c" ?>
4
5 <!DOCTYPE html>
6
7 <html>
8 <head>
9
10 <title>Employee List</title>
11
12 <link rel="stylesheet"
13   href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
14
15 </head>
16
17 <body>
18
19 <div class="container mt-4">
20
21   <h2>Employee List</h2>
22
23   <a href="new"
24     class="btn btn-success mb-3">
25     Add Employee
26   </a>
27
28   <table class="table table-bordered">
29
30     <tr>
31       <th>ID</th>
32       <th>Name</th>
33       <th>Email</th>
34       <th>Position</th>
35       <th>Action</th>
36     </tr>

```

```

38 <c:forEach var="employee"
39     items="${listEmployee}">
40
41 <tr>
42
43 <td>${employee.id}</td>
44 <td>${employee.name}</td>
45 <td>${employee.email}</td>
46 <td>${employee.position}</td>
47
48 <td>
49
50 <a href="edit?id=${employee.id}"
51     class="btn btn-warning btn-sm">
52     Edit
53 </a>
54
55 <a href="delete?id=${employee.id}"
56     class="btn btn-danger btn-sm">
57     Delete
58 </a>
59
60 </td>
61 </tr>
62 </tr>
63
64 </c:forEach>
65
66 </table>
67
68 </div>
69
70 </body>
71 </html>

```

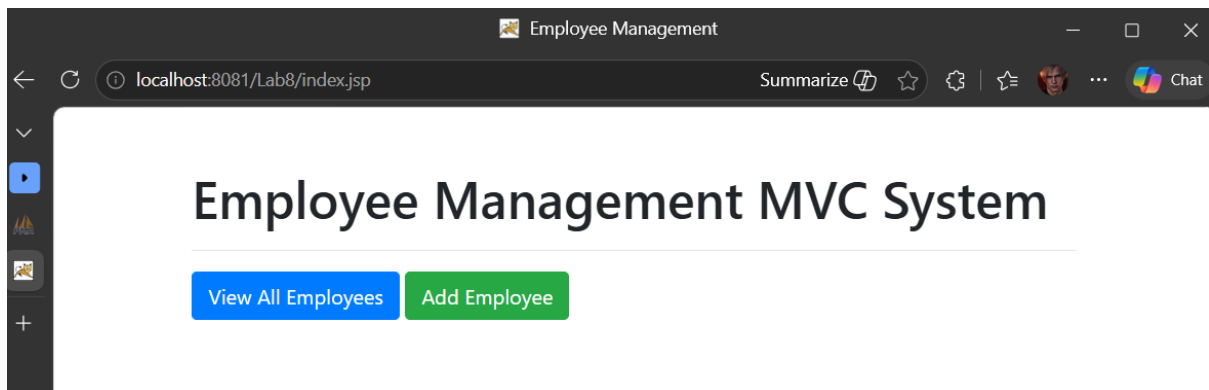
```

...ml index.html x Employee.java x EmployeeDAO.java x EmployeeServlet.java x index.jsp x employeeForm.jsp x employeeList.jsp x error.jsp x
Source History
1 <%@ page isErrorPage="true" %>
2
3 <!DOCTYPE html>
4
5 <html>
6
7 <head>
8 <title>Error</title>
9 </head>
10
11 <body>
12
13 <h1>Error</h1>
14
15 <h3>
16 <%= exception.getMessage() %>
17 </h3>
18
19 </body>
20
21 </html>

```

```
...ml index.html x Employee.java x web.xml x EmployeeDAO.java x EmployeeServlet.java x index.  
Source History C:\Users\User\Documents\TAHUN 2 SEM 4\LAB8_s76276\La  
1 <?xml version="1.0" encoding="UTF-8"?>  
2  
3 <web-app  
4 xmlns="https://jakarta.ee/xml/ns/jakartaee"  
5 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
6 xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee  
7 https://jakarta.ee/xml/ns/jakartaee/web-app_5_0.xsd"  
8 version="5.0">  
9  
10 <welcome-file-list>  
11 <welcome-file>index.jsp</welcome-file>  
12 </welcome-file-list>  
13  
14 </web-app>
```

Output:



The screenshot shows a web browser window titled "Employee Form". The address bar displays "localhost:8081/Lab8/new". The main content area is titled "Add Employee" and contains three input fields labeled "Name", "Email", and "Position". Below the input fields are two buttons: "Save" (green) and "Back" (grey). A vertical sidebar on the left contains navigation icons, including a play button, a folder icon, and a plus sign.

Employee List

localhost:8081/Lab8/list

☆ ⚙️ ⌵ 👤 ... Chat

Employee List

Add Employee

ID	Name	Email	Position	Action
1	NORSYAFIQAA BINTI MOHD HISHAM	s77777@ocean.umat.edu.my	secretary	Edit Delete

Employee List

localhost:8081/Lab8/list

☆ ⚙️ ⌵ 👤 ... Chat

Employee List

Add Employee

1	NORSYAFIQAA BINTI MOHD HISHAM	s77777@ocean.umat.edu.my	secretary	Edit Delete
2	MUHAMMAD SYAMIL IRFAN BIN MOHD HISHAM	s88888@ocean.umat.edu.my	manager	Edit Delete

Edit Action

Employee List

localhost:8081/Lab8/list

Employee List

Add Employee

ID	Name	Email	Position	Action
1	NORSYAFIQAA BINTI MOHD HISHAM	s77777@ocean.umt.edu.my	secretary	Edit Delete
2	MUHAMMAD SYAMIL IRFAN BIN MOHD HISHAM	s88888@ocean.umt.edu.my	student	Edit Delete

Delete action:

Employee List

localhost:8081/Lab8/list

Employee List

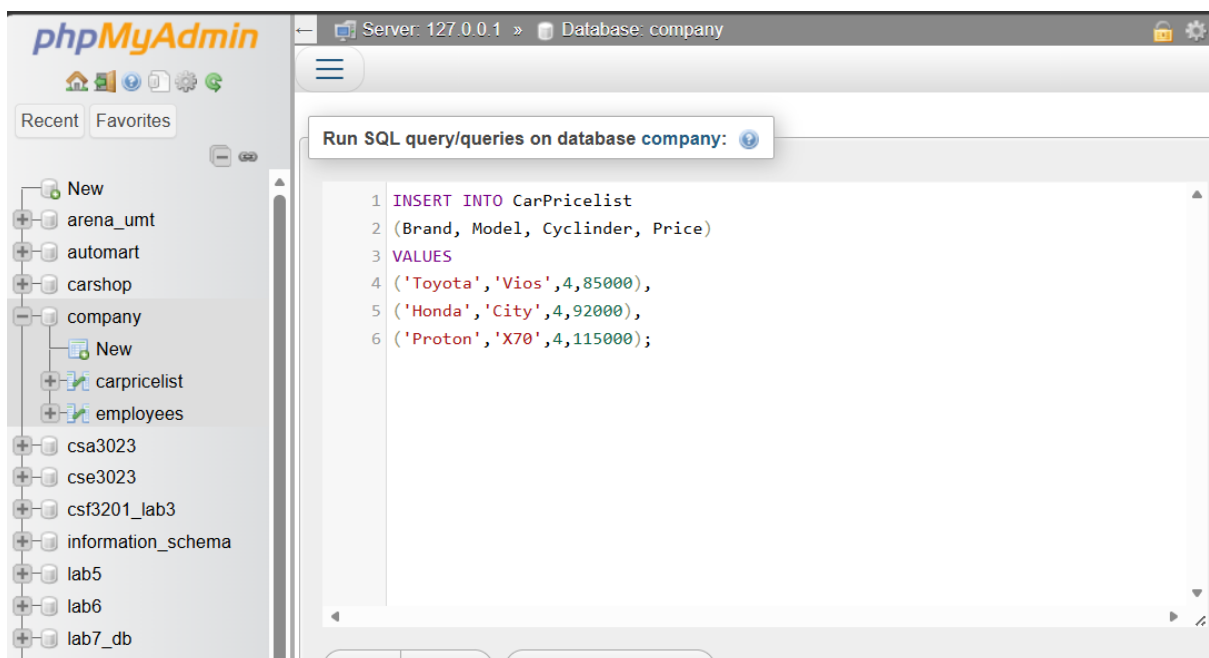
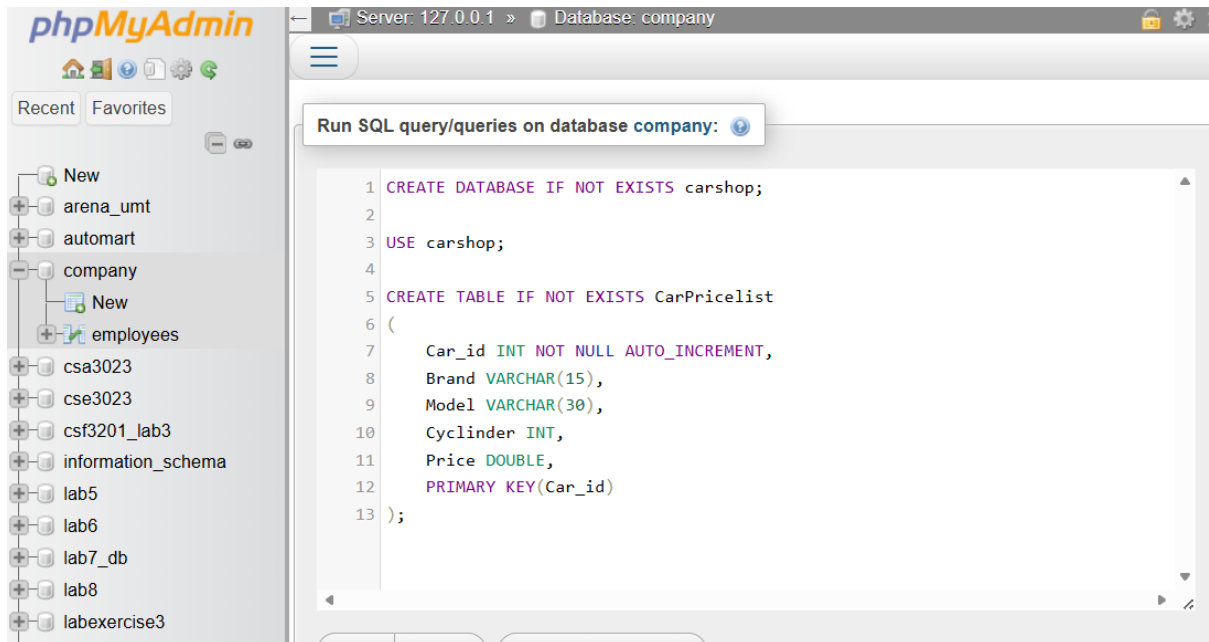
Add Employee

ID	Name	Email	Position	Action
2	MUHAMMAD SYAMIL IRFAN BIN MOHD HISHAM	s88888@ocean.umt.edu.my	student	Edit Delete

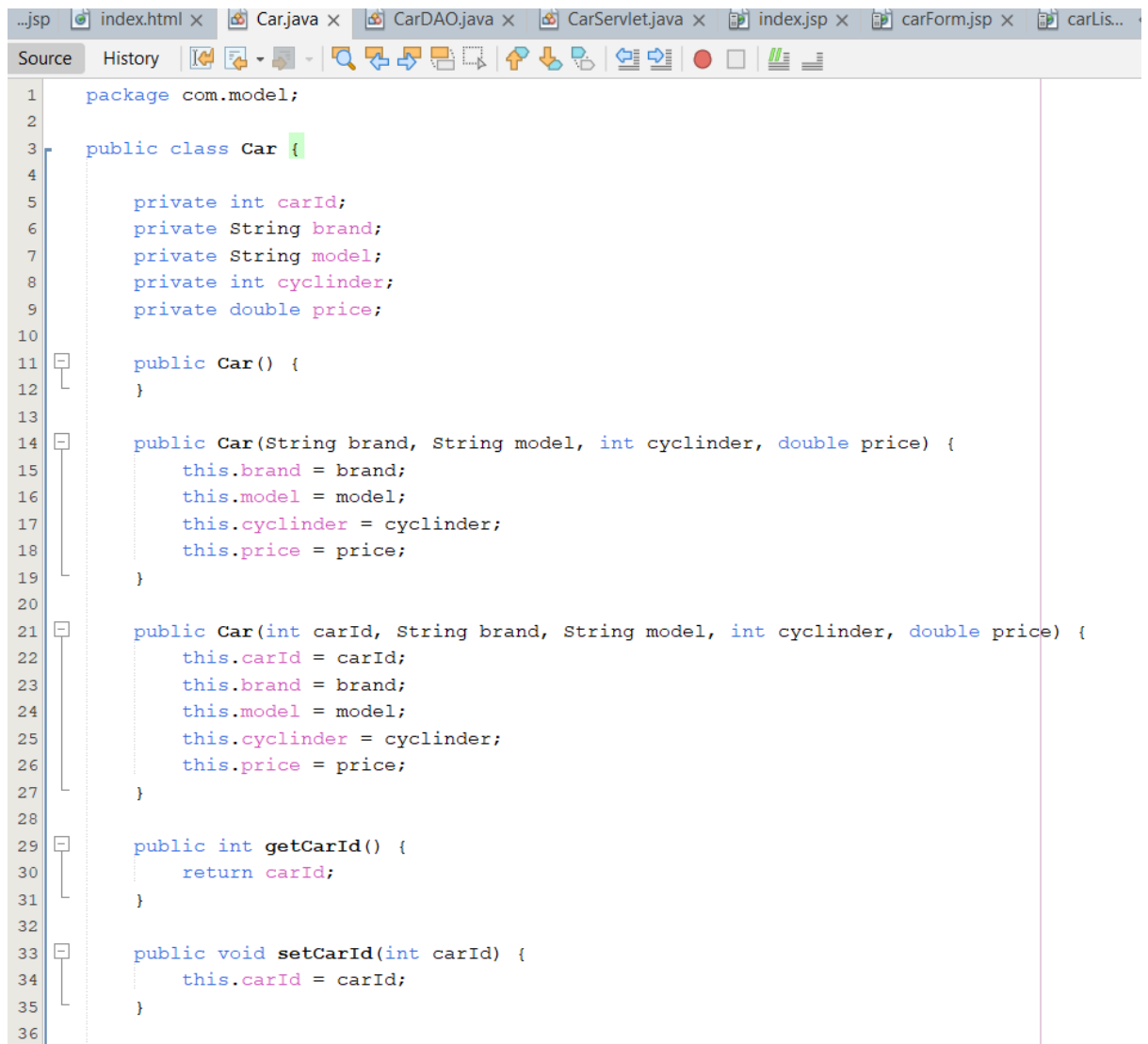
Exercise

Implement MVC for Car Shop

1. Using this database schema, create an MVC Application [CRUD] for Car Shop, using JSP, Servlet, and MySQL.



2. The application should be able to handle these activities:
 - View all data
 - Add new data
 - Edit/Update current data
 - Delete the specific data



```
1 package com.model;
2
3 public class Car {
4
5     private int carId;
6     private String brand;
7     private String model;
8     private int cylinder;
9     private double price;
10
11     public Car() {
12     }
13
14     public Car(String brand, String model, int cylinder, double price) {
15         this.brand = brand;
16         this.model = model;
17         this.cylinder = cylinder;
18         this.price = price;
19     }
20
21     public Car(int carId, String brand, String model, int cylinder, double price) {
22         this.carId = carId;
23         this.brand = brand;
24         this.model = model;
25         this.cylinder = cylinder;
26         this.price = price;
27     }
28
29     public int getCarId() {
30         return carId;
31     }
32
33     public void setCarId(int carId) {
34         this.carId = carId;
35     }
36 }
```

```
37  [-] public String getBrand() {  
38      return brand;  
39  }  
40  
41  [-] public void setBrand(String brand) {  
42      this.brand = brand;  
43  }  
44  
45  [-] public String getModel() {  
46      return model;  
47  }  
48  
49  [-] public void setModel(String model) {  
50      this.model = model;  
51  }  
52  
53  [-] public int getCyclinder() {  
54      return cyclinder;  
55  }  
56  
57  [-] public void setCyclinder(int cyclinder) {  
58      this.cyclinder = cyclinder;  
59  }  
60  
61  [-] public double getPrice() {  
62      return price;  
63  }  
64  
65  [-] public void setPrice(double price) {  
66      this.price = price;  
67  }  
68  }
```

```
...jsp index.html x Car.java x CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carLis... < > v
Source History
1 package com.dao;
2
3 import com.model.Car;
4 import java.sql.*;
5 import java.util.ArrayList;
6 import java.util.List;
7
8 public class CarDAO {
9
10     private String jdbcURL = "jdbc:mysql://localhost:3306/carshop";
11     private String jdbcUsername = "root";
12     private String jdbcPassword = "admin";
13
14     private static final String INSERT_CAR_SQL =
15         "INSERT INTO CarPricelist (Brand, Model, Cyclinder, Price) VALUES (?, ?, ?, ?)";
16
17     private static final String SELECT_CAR_BY_ID =
18         "SELECT * FROM CarPricelist WHERE Car_id=?";
19
20     private static final String SELECT_ALL_CARS =
21         "SELECT * FROM CarPricelist";
22
23     private static final String DELETE_CAR_SQL =
24         "DELETE FROM CarPricelist WHERE Car_id=?";
25
26     private static final String UPDATE_CAR_SQL =
27         "UPDATE CarPricelist SET Brand=?, Model=?, Cyclinder=?, Price=? WHERE Car_id=?";
28
29     protected Connection getConnection() {
30
31         Connection connection = null;
32
33         try {
34
35             Class.forName("com.mysql.cj.jdbc.Driver");
36
```

```

37         connection = DriverManager.getConnection(
38             jdbcURL,
39             jdbcUsername,
40             jdbcPassword);
41
42     } catch (Exception e) {
43         e.printStackTrace();
44     }
45
46     return connection;
47 }
48
49 public void insertCar(Car car) throws SQLException {
50
51     try (Connection connection = getConnection();
52         PreparedStatement statement =
53             connection.prepareStatement(INSERT_CAR_SQL)) {
54
55         statement.setString(1, car.getBrand());
56         statement.setString(2, car.getModel());
57         statement.setInt(3, car.getCylinder());
58         statement.setDouble(4, car.getPrice());
59
60         statement.executeUpdate();
61     }
62 }
63
64 public Car selectCar(int id) {
65
66     Car car = null;
67
68     try (Connection connection = getConnection();
69         PreparedStatement statement =
70             connection.prepareStatement(SELECT_CAR_BY_ID)) {
71
72         statement.setInt(1, id);

```

```

73
74         ResultSet rs = statement.executeQuery();
75
76         while (rs.next()) {
77
78             String brand = rs.getString("Brand");
79             String model = rs.getString("Model");
80             int cylcylinder = rs.getInt("Cylcylinder");
81             double price = rs.getDouble("Price");
82
83             car = new Car(id, brand, model, cylcylinder, price);
84         }
85
86     } catch (Exception e) {
87         e.printStackTrace();
88     }
89
90     return car;
91 }
92
93 public List<Car> selectAllCars() {
94
95     List<Car> cars = new ArrayList<>();
96
97     try (Connection connection = getConnection();
98         PreparedStatement statement =
99             connection.prepareStatement(SELECT_ALL_CARS)) {
100
101         ResultSet rs = statement.executeQuery();
102
103         while (rs.next()) {
104
105             cars.add(new Car(
106                 rs.getInt("Car_id"),
107                 rs.getString("Brand"),
108                 rs.getString("Model"),

```

```

109         rs.getInt("Cyclinder"),
110         rs.getDouble("Price")));
111     }
112
113     } catch (Exception e) {
114         e.printStackTrace();
115     }
116
117     return cars;
118 }
119
120 public boolean updateCar(Car car) throws SQLException {
121
122     boolean rowUpdated;
123
124     try (Connection connection = getConnection();
125         PreparedStatement statement =
126             connection.prepareStatement(UPDATE_CAR_SQL)) {
127
128         statement.setString(1, car.getBrand());
129         statement.setString(2, car.getModel());
130         statement.setInt(3, car.getCyclinder());
131         statement.setDouble(4, car.getPrice());
132         statement.setInt(5, car.getCarId());
133
134         rowUpdated = statement.executeUpdate() > 0;
135     }
136
137     return rowUpdated;
138 }
139
140 public boolean deleteCar(int id) throws SQLException {
141
142     boolean rowDeleted;
143
144     try (Connection connection = getConnection();
145         PreparedStatement statement =
146             connection.prepareStatement(DELETE_CAR_SQL)) {
147
148         statement.setInt(1, id);
149
150         rowDeleted = statement.executeUpdate() > 0;
151     }
152
153     return rowDeleted;
154 }
155 }

```

```
..jsp index.html x Car.java x CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carLis... < >
Source History
1 package com.web;
2
3 import com.dao.CarDAO;
4 import com.model.Car;
5
6 import java.io.IOException;
7 import java.sql.SQLException;
8 import java.util.List;
9
10 import javax.servlet.RequestDispatcher;
11 import javax.servlet.ServletException;
12 import javax.servlet.annotation.WebServlet;
13 import javax.servlet.http.*;
14
15 @WebServlet("/")
16 public class CarServlet extends HttpServlet {
17
18     private CarDAO carDAO;
19
20     public void init() {
21         carDAO = new CarDAO();
22     }
23
24     protected void doPost(HttpServletRequest request,
25                             HttpServletResponse response)
26         throws ServletException, IOException {
27
28         doGet(request, response);
29     }
30
31     protected void doGet(HttpServletRequest request,
32                             HttpServletResponse response)
33         throws ServletException, IOException {
34
35         String action = request.getServletPath();
36     }
37 }
```



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71

try {

    switch (action) {

        case "/new":
            showNewForm(request, response);
            break;

        case "/insert":
            insertCar(request, response);
            break;

        case "/delete":
            deleteCar(request, response);
            break;

        case "/edit":
            showEditForm(request, response);
            break;

        case "/update":
            updateCar(request, response);
            break;

        default:
            listCars(request, response);
            break;

    }

} catch (SQLException ex) {
    throw new ServletException(ex);
}

private void listCars(HttpServletRequest request,
```

```

72         HttpServletResponse response)
73         throws SQLException, IOException, ServletException {
74
75         List<Car> listCars = carDAO.selectAllCars();
76
77         request.setAttribute("listCars", listCars);
78
79         RequestDispatcher dispatcher =
80             request.getRequestDispatcher("carList.jsp");
81
82         dispatcher.forward(request, response);
83     }
84
85     private void showNewForm(HttpServletRequest request,
86                             HttpServletResponse response)
87         throws ServletException, IOException {
88
89         RequestDispatcher dispatcher =
90             request.getRequestDispatcher("carForm.jsp");
91
92         dispatcher.forward(request, response);
93     }
94
95     private void showEditForm(HttpServletRequest request,
96                               HttpServletResponse response)
97         throws SQLException, ServletException, IOException {
98
99         int id = Integer.parseInt(request.getParameter("id"));
100
101         Car existingCar = carDAO.selectCar(id);
102
103         RequestDispatcher dispatcher =
104             request.getRequestDispatcher("carForm.jsp");
105
106         request.setAttribute("car", existingCar);
107

```

```

108     dispatcher.forward(request, response);
109 }
110
111 private void insertCar(HttpServletRequest request,
112                       HttpServletResponse response)
113     throws SQLException, IOException {
114
115     String brand = request.getParameter("brand");
116     String model = request.getParameter("model");
117     int cylcylinder = Integer.parseInt(request.getParameter("cylcylinder"));
118     double price = Double.parseDouble(request.getParameter("price"));
119
120     Car newCar = new Car(brand, model, cylcylinder, price);
121
122     carDAO.insertCar(newCar);
123
124     response.sendRedirect("list");
125 }
126
127 private void updateCar(HttpServletRequest request,
128                       HttpServletResponse response)
129     throws SQLException, IOException {
130
131     int id = Integer.parseInt(request.getParameter("id"));
132
133     String brand = request.getParameter("brand");
134     String model = request.getParameter("model");
135     int cylcylinder = Integer.parseInt(request.getParameter("cylcylinder"));
136     double price = Double.parseDouble(request.getParameter("price"));
137
138     Car car = new Car(id, brand, model, cylcylinder, price);
139
140     carDAO.updateCar(car);
141
142     response.sendRedirect("list");
143 }
144
145 private void deleteCar(HttpServletRequest request,
146                       HttpServletResponse response)
147     throws SQLException, IOException {
148
149     int id = Integer.parseInt(request.getParameter("id"));
150
151     carDAO.deleteCar(id);
152
153     response.sendRedirect("list");
154 }
155 }

```

```
...va CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carList.jsp x web.xml x
Source History
<%@page contentType="text/html" pageEncoding="UTF-8"%>
2
3 <!DOCTYPE html>
4 <html>
5 <head>
6 <title>Car Shop Management</title>
7 </head>
8
9 <body>
10
11 <h1>Car Shop Management System</h1>
12
13 <ul>
14 <li><a href="list">View All Cars</a></li>
15 <li><a href="new">Add New Car</a></li>
16 </ul>
17
18 </body>
</html>
```

```
...va CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carList.jsp x web.xml x
Source History
<%@ page contentType="text/html; charset=UTF-8"%>
2 <%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
3
4 <html>
5 <head>
6 <title>Car Form</title>
7 </head>
8
9 <body>
10
11 <c:if test="${car != null}">
12 <form action="update" method="post">
13 </c:if>
14
15 <c:if test="${car == null}">
16 <form action="insert" method="post">
17 </c:if>
18
19 <c:if test="${car != null}">
20 <input type="hidden" name="id" value="${car.carId}">
21 </c:if>
22
23 Brand :
24 <input type="text" name="brand" value="${car.brand}">
25 <br><br>
26
27 Model :
28 <input type="text" name="model" value="${car.model}">
29 <br><br>
30
31 Cyclinder :
32 <input type="number" name="cyclinder" value="${car.cyclinder}">
33 <br><br>
34
35 Price :
36 <input type="number" step="0.01" name="price" value="${car.price}">
```

```

37     <br><br>
38
39     <input type="submit" value="Save">
40
41 </form>
42
43 </body>
44 </html>

```

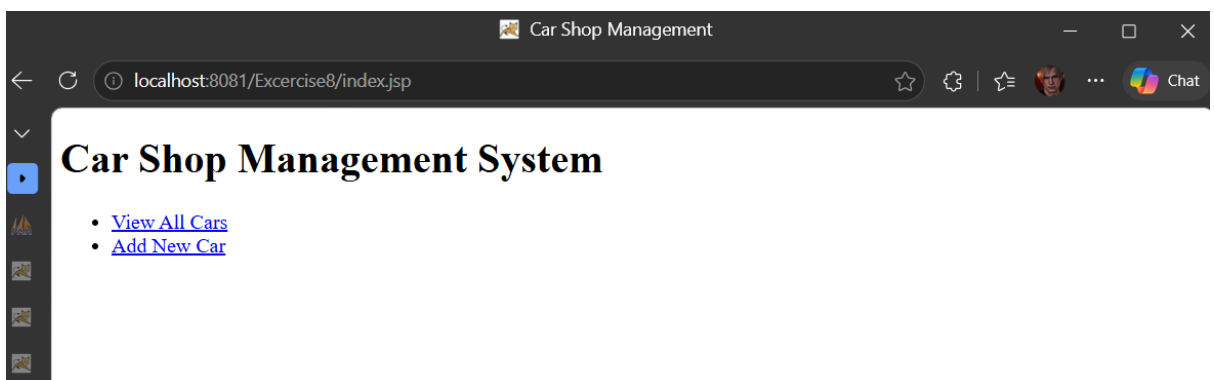
```

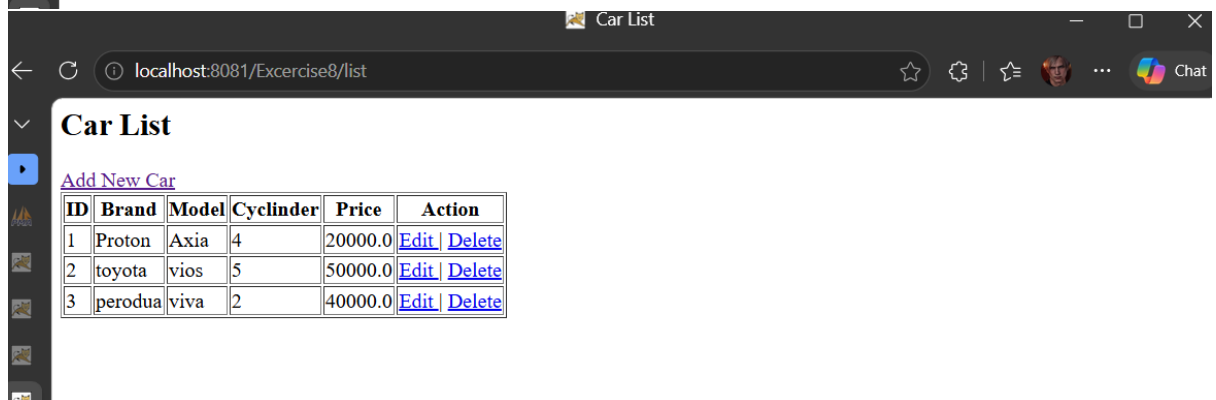
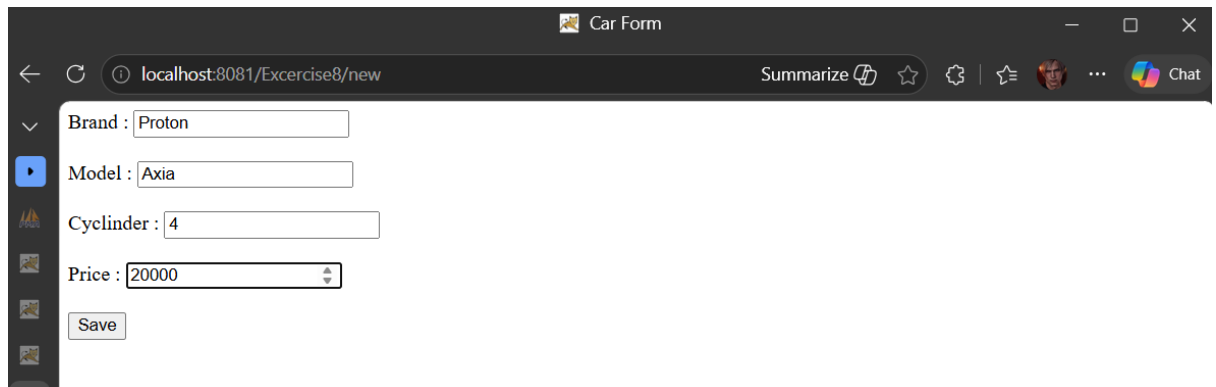
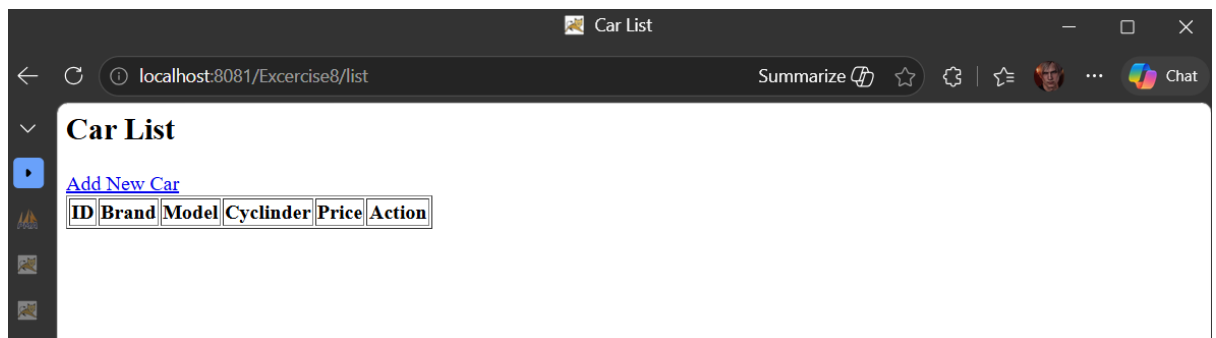
...va CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carList.jsp x web.xml x
Source History
<%@ page contentType="text/html; charset=UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<html>
<head>
<title>Car List</title>
</head>
<body>
<h2>Car List</h2>
<a href="new">Add New Car</a>
<table border="1">
<tr>
<th>ID</th>
<th>Brand</th>
<th>Model</th>
<th>Cyclinder</th>
<th>Price</th>
<th>Action</th>
</tr>
<c:forEach var="car" items="${listCars}">
<tr>
<td>${car.carId}</td>
<td>${car.brand}</td>
<td>${car.model}</td>
<td>${car.cyclinder}</td>
<td>${car.price}</td>
<td>

```

```
37
38 <a href="edit?id=${car.carId}">
39   Edit
40 </a>
41
42 |
43
44 <a href="delete?id=${car.carId}">
45   Delete
46 </a>
47
48 </td>
49
50 </tr>
51
52 </c:forEach>
53
54 </table>
55
56 </body>
57 </html>
```

```
...va CarDAO.java x CarServlet.java x index.jsp x carForm.jsp x carList.jsp x web.xml x
Source History
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
4         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5         xsi:schemaLocation="
6             http://xmlns.jcp.org/xml/ns/javaee
7             http://xmlns.jcp.org/xml/ns/javaee/web-app 4.0.xsd"
8         version="4.0">
9
10 <welcome-file-list>
11 <welcome-file>index.jsp</welcome-file>
12 </welcome-file-list>
13
14 </web-app>
```





Edit action:

Car Form

localhost:8081/Exercise8/edit?id=1

Brand :

Model :

Cyclinder :

Price :

After edit:

Car List

localhost:8081/Exercise8/list

[Add New Car](#)

ID	Brand	Model	Cyclinder	Price	Action
1	Proton	Axia	2	20000.0	Edit Delete
2	toyota	vios	5	50000.0	Edit Delete
3	perodua	viva	2	40000.0	Edit Delete

After delete:

Car List

localhost:8081/Exercise8/list

[Add New Car](#)

ID	Brand	Model	Cyclinder	Price	Action
1	Proton	Axia	2	20000.0	Edit Delete
3	perodua	viva	2	40000.0	Edit Delete